Year 10 · Chapter 2 · Measurement: answers (teacher copy)

2.1 Converting units (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 4 × 100 = 400 cm
2. Ex 1b: 7 × 10 = 70 mm
3. Ex 1c: 3 000 ÷ 1 000 = 3 km
4. Ex 1d: 250 ÷ 100 = 2.5 m
5. Ex 2a: 3 × 1 000 = 3 000 g
6. Ex 2b: 2 500 ÷ 1 000 = 2.5 L
7. Ex 2c: 5 × 60 = 300 seconds
8. Ex 2d: 2 × 60 = 120 minutes, 120 × 60 = 7 200 seconds

Easy

1. 150 mm
2. 200 cm
3. 12 000 m
4. 300 mm
5. 2.5 cm
6. 6 m
7. 0.4 km
8. 3 cm
9. 180 cm
10. 0.45 m
11. 2 500 m
12. 15 cm

Medium

1. 2 000 g
2. 0.75 kg
3. 3 000 kg
4. 1.5 t
5. 1 500 mL
6. 0.5 L
7. 0.375 L
8. 9 000 mL
9. 240 seconds
10. 3 minutes
11. 72 hours
12. 86 400 seconds
13. 5.3 × 1 000 = 5 300 m; 5 300 − 2 800 = 2 500 m
14. 2 × 1 000 = 2 000 mL; 2 000 ÷ 250 = 8 glasses

2.2 Perimeter (booklet p. 5)

WE DO examples (teacher)

1. Ex 1a: 8 + 6 + 5 = 19 cm
2. Ex 1b: 4 × 6 = 24 cm
3. Ex 1c: 6 × 3 = 18 m
4. Ex 1d: 9 + 4 + 9 + 4 = 26 m
5. Ex 2a: a = 8 − 3 = 5 m
6. Ex 2b: b = 10 − 4 = 6 m
7. Ex 2c: P = 4 + 5 + 6 + 3 + 10 + 8 = 36 m
8. Ex 2d: 36 × $20 = $720

Easy

1. 7 + 6 + 4 = 17 cm
2. 12 + 7 + 8 + 9 = 36 m
3. 6 + 5 + 9 + 4 = 24 mm
4. 10 + 4 + 6 + 6 + 4 = 30 cm
5. 4 × 8 = 32 cm
6. 3 × 7 = 21 m
7. 5 × 4 = 20 cm
8. 4 × 6 = 24 mm
9. 6 + 3 + 6 + 3 = 18 cm
10. 5 + 9 + 5 + 9 = 28 m
11. 6 + 8 + 8 = 22 cm
12. 12 + 5 + 12 + 5 = 34 mm

Medium

1. 4 × 2 = 8 cm
2. 7 + 5 + 7 + 5 = 24 cm
3. 4.5 + 2 + 4.5 + 2 = 13 m
4. 9 + 6 + 9 + 6 = 30 mm
5. 12 − 5 = 7 m
6. 9 − 4 = 5 cm
7. 5 + 4 = 9 m
8. 3 + 4 = 7 cm
9. ? = 4 m, ? = 5 m; P = 4 + 4 + 5 + 2 + 9 + 6 = 30 m
10. ? = 2 cm, ? = 10 cm; P = 4 + 4 + 6 + 2 + 10 + 6 = 32 cm
11. ? = 7 m, ? = 6 m; P = 3 + 4 + 7 + 4 + 3 + 6 + 13 + 6 = 46 m
12. P = 50 + 30 + 50 + 30 = 160 m, 160 × $12 = $1 920
13. one lap = 4 × 200 = 800 m, 3 × 800 = 2 400 m

2.3 Circumference (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: d = 2 × 6 = 12 cm
2. Ex 1b: r = 14 ÷ 2 = 7 m
3. Ex 1c: C = π × 10 = 31.4 cm
4. Ex 1d: C = 2 × π × 4 = 25.1 m
5. Ex 2a: C = π × 6 = 18.8 cm (not 2 × π × 6: 6 cm is the diameter)
6. Ex 2b: ½ × π × 10 + 10 = 15.7 + 10 = 25.7 cm
7. Ex 2c: ¼ × 2 × π × 4 + 4 + 4 = 6.3 + 8 = 14.3 m
8. Ex 2d: 12 + 12 + 6 + ½ × π × 6 = 30 + 9.4 = 39.4 cm

Easy

1. d = 10 cm
2. r = 4 cm
3. d = 18 m
4. r = 10 m
5. 12.6 cm
6. 28.3 m
7. 62.8 cm
8. 47.1 mm
9. 37.7 m
10. 50.3 cm
11. 69.1 mm
12. 157.1 cm

Medium

1. 12.6 + 8 = 20.6 cm
2. d = 6; 9.4 + 6 = 15.4 m
3. 31.4 + 20 = 51.4 mm
4. d = 18; 28.3 + 18 = 46.3 cm
5. 3.1 + 4 = 7.1 cm
6. 9.4 + 12 = 21.4 m
7. 15.7 + 20 = 35.7 cm
8. 12.6 + 16 = 28.6 mm
9. 24 + 6.3 = 30.3 cm
10. 38 + 12.6 = 50.6 m
11. 50 + 15.7 = 65.7 cm
12. 14 + 3.1 = 17.1 m
13. 2 × π × 3 = 18.8 m
14. π × 60 = 188.5 cm
15. 3.1 + 2 = 5.1 m

2.4 Area (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 8 × 5 = 40 cm²
2. Ex 1b: 6 × 6 = 36 m²
3. Ex 1c: ½ × 10 × 4 = 20 cm²
4. Ex 1d: ½ × 12 × 7 = 42 m²
5. Ex 2a: 9 × 4 = 36 cm²
6. Ex 2b: 10 × 6 = 60 m² (not 7)
7. Ex 2c: ½ × (4 + 6) × 6 = 30 cm²
8. Ex 2d: ½ × 10 × 6 = 30 cm²

Easy

1. 24 cm²
2. 30 m²
3. 16 cm²
4. 81 m²
5. 24 cm²
6. 25 m²
7. 18 cm²
8. 18 mm²
9. 30 cm²
10. 20 m²
11. 28 cm²
12. 27 mm²

Medium

1. 21 cm²
2. 60 m²
3. 64 mm²
4. 60 cm²
5. 20 cm²
6. 42 m²
7. 24 cm²
8. 25 mm²
9. 24 cm²
10. 24 m²
11. 50 mm²
12. 14 cm²
13. 96 m², $1 440
14. 7.5 m²

2.5 Area of circles (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: π × 5² = 78.5 cm²
2. Ex 1b: π × 2² = 12.6 m²
3. Ex 1c: r = 8 ÷ 2 = 4, π × 4² = 50.3 cm²
4. Ex 1d: r = 12 ÷ 2 = 6, π × 6² = 113.1 m²
5. Ex 2a: r = 5, ½ × π × 5² = 39.3 cm²
6. Ex 2b: ¼ × π × 4² = 12.6 m²
7. Ex 2c: 12 × 6 + ½ × π × 3² = 72 + 14.1 = 86.1 cm²
8. Ex 2d: ½ × π × 6² = 56.5 m²

Easy

1. $\pi \times 3^2 = 28.3\ m^2$
2. $\pi \times 4^2 = 50.3\ cm^2$
3. $\pi \times 7^2 = 153.9\ mm^2$
4. $\pi \times 10^2 = 314.2\ cm^2$
5. $r = 2 \div 2 = 1$, $\pi \times 1^2 = 3.1\ cm^2$
6. $r = 14 \div 2 = 7$, $\pi \times 7^2 = 153.9\ m^2$
7. $r = 20 \div 2 = 10$, $\pi \times 10^2 = 314.2\ cm^2$
8. $r = 18 \div 2 = 9$, $\pi \times 9^2 = 254.5\ mm^2$
9. $\pi \times 1^2 = 3.1\ m^2$
10. $r = 4 \div 2 = 2$, $\pi \times 2^2 = 12.6\ cm^2$
11. $\pi \times 6^2 = 113.1\ mm^2$
12. $r = 30 \div 2 = 15$, $\pi \times 15^2 = 706.9\ cm^2$

Medium

1. $r = 3$, $\frac{1}{2} \times \pi \times 3^2 = 14.1\ cm^2$
2. $\frac{1}{2} \times \pi \times 4^2 = 25.1\ m^2$
3. $r = 8$, $\frac{1}{2} \times \pi \times 8^2 = 100.5\ mm^2$
4. $\frac{1}{2} \times \pi \times 7^2 = 77.0\ cm^2$
5. $\frac{1}{4} \times \pi \times 1^2 = 0.8\ m^2$
6. $\frac{1}{4} \times \pi \times 3^2 = 7.1\ cm^2$
7. $\frac{1}{4} \times \pi \times 12^2 = 113.1\ cm^2$
8. $\frac{1}{4} \times \pi \times 5^2 = 19.6\ mm^2$
9. $8 \times 4 + \frac{1}{2} \times \pi \times 2^2 = 32 + 6.3 = 38.3\ cm^2$
10. $10 \times 6 + \frac{1}{2} \times \pi \times 3^2 = 60 + 14.1 = 74.1\ m^2$
11. $16 \times 8 + \frac{1}{2} \times \pi \times 4^2 = 128 + 25.1 = 153.1\ cm^2$
12. $5 \times 2 + \frac{1}{2} \times \pi \times 1^2 = 10 + 1.6 = 11.6\ m^2$
13. $r = 15$, $\pi \times 15^2 = 706.9\ cm^2$
14. $\pi \times 1.5^2 = 7.1\ m^2$
15. $\pi \times 4^2 = 50.3\ m^2$

2.6 Surface area (booklet p. 13)

WE DO examples (teacher)

1. Ex 1a: $6 \times 3 \times 3 = 54\ cm^2$
2. Ex 1b: $6 \times 5 \times 5 = 150\ m^2$
3. Ex 1c: $2 \times 15 + 2 \times 10 + 2 \times 6 = 62\ cm^2$
4. Ex 1d: $2 \times 16 + 2 \times 40 + 2 \times 10 = 132\ m^2$
5. Ex 2a: 5 faces: 2 triangles and 3 rectangles
6. Ex 2b: $\frac{1}{2} \times 4 \times 3 = 6\ m^2$
7. Ex 2c: $6 \times 4 + 6 \times 3 + 6 \times 5 = 72\ m^2$
8. Ex 2d: $2 \times 6 + 6 \times 4 + 6 \times 3 + 6 \times 5 = 84\ m^2$

Easy

1. $6 \times 2 \times 2 = 24\ cm^2$
2. $6 \times 4 \times 4 = 96\ m^2$
3. $6 \times 10 \times 10 = 600\ mm^2$
4. $6 \times 6 \times 6 = 216\ cm^2$
5. $2 \times 20 + 2 \times 40 + 2 \times 8 = 136\ cm^2$
6. $2 \times 12 + 2 \times 18 + 2 \times 6 = 72\ m^2$
7. $2 \times 25 + 2 \times 10 + 2 \times 10 = 90\ cm^2$
8. $2 \times 7 + 2 \times 14 + 2 \times 2 = 46\ m^2$
9. $2 \times 36 + 2 \times 60 + 2 \times 15 = 222\ cm^2$
10. $2 \times 36 + 2 \times 36 + 2 \times 16 = 176\ mm^2$
11. $2 \times 100 + 2 \times 200 + 2 \times 50 = 700\ cm^2$
12. $2 \times 9 + 2 \times 24 + 2 \times 24 = 114\ m^2$

Medium

1. $2 \times 6 + 10 \times 4 + 10 \times 3 + 10 \times 5 = 132\ cm^2$
2. $2 \times 24 + 5 \times 8 + 5 \times 6 + 5 \times 10 = 168\ m^2$
3. $2 \times 30 + 2 \times 12 + 2 \times 5 + 2 \times 13 = 120\ cm^2$
4. $2 \times 54 + 20 \times 12 + 20 \times 9 + 20 \times 15 = 828\ m^2$
5. $2 \times 12 + 10 \times 6 + 10 \times 5 + 10 \times 5 = 184\ cm^2$
6. $2 \times 12 + 6 \times 8 + 6 \times 5 + 6 \times 5 = 132\ m^2$
7. $2 \times 60 + 4 \times 10 + 4 \times 13 + 4 \times 13 = 264\ cm^2$
8. $2 \times 48 + 5 \times 16 + 5 \times 10 + 5 \times 10 = 276\ mm^2$
9. $2 \times 5 + 2 \times 10 + 2 \times 8 = 46\ m^2$
10. $2 \times 24 + 3 \times 6 + 3 \times 8 + 3 \times 10 = 120\ cm^2$
11. $2 \times 1.5 + 2 \times 3 + 2 \times 2 = 13\ m^2$
12. $2 \times 30 + 10 \times 5 + 10 \times 12 + 10 \times 13 = 360\ mm^2$
13. $6 \times 20 \times 20 = 2\,400\ cm^2$
14. $2 \times 1\,500 + 2 \times 1\,000 + 2 \times 600 = 6\,200\ cm^2$

2.7 Volume and capacity (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: $2 \times 2 \times 2 = 8\ m^3$
2. Ex 1b: $5 \times 3 \times 2 = 30\ cm^3$
3. Ex 1c: $10 \times 2 \times 4 = 80\ m^3$
4. Ex 1d: $6 \times 4 \times 5 = 120\ mm^3$
5. Ex 2a: $\frac{1}{2} \times 4 \times 3 \times 10 = 6 \times 10 = 60\ cm^3$
6. Ex 2b: $\pi \times 2^2 \times 6 = 75.4\ cm^3$
7. Ex 2c: $1\ cm^3 = 1\ mL$, so 75.4 mL
8. Ex 2d: $2\,000 \div 1\,000 = 2\ L$

Easy

1. $3 \times 3 \times 3 = 27\ cm^3$
2. $5 \times 5 \times 5 = 125\ m^3$
3. $10 \times 10 \times 10 = 1\,000\ cm^3$
4. $4 \times 4 \times 4 = 64\ mm^3$
5. $6 \times 4 \times 3 = 72\ cm^3$
6. $8 \times 5 \times 2 = 80\ m^3$
7. $10 \times 10 \times 3 = 300\ cm^3$
8. $7 \times 2 \times 5 = 70\ mm^3$
9. $12 \times 5 \times 4 = 240\ cm^3$
10. $20 \times 6 \times 10 = 1\,200\ cm^3$
11. $9 \times 3 \times 2 = 54\ m^3$
12. $2.5 \times 2 \times 4 = 20\ m^3$

Medium

1. $\frac{1}{2} \times 6 \times 8 \times 5 = 24 \times 5 = 120\ cm^3$
2. $\frac{1}{2} \times 5 \times 12 \times 10 = 30 \times 10 = 300\ m^3$
3. $\frac{1}{2} \times 4 \times 4 \times 2 = 8 \times 2 = 16\ cm^3$
4. $\frac{1}{2} \times 12 \times 9 \times 20 = 54 \times 20 = 1\,080\ m^3$
5. $\pi \times 3^2 \times 10 = 282.7\ cm^3$
6. $\pi \times 1^2 \times 2 = 6.3\ m^3$
7. $\pi \times 5^2 \times 4 = 314.2\ cm^3$
8. $r = 5$, $\pi \times 5^2 \times 2 = 157.1\ m^3$
9. $20 \times 20 \times 10 = 4\,000\ cm^3 = 4\ L$
10. $50 \times 20 \times 10 = 10\,000\ cm^3 = 10\ L$
11. $30 \times 10 \times 10 = 3\,000\ cm^3 = 3\ L$
12. $25 \times 20 \times 10 = 5\,000\ cm^3 = 5\ L$
13. $r = 4$, $\pi \times 4^2 \times 2 = 100.5\ m^3$, about 100 531 L
14. $60 \times 30 \times 40 = 72\,000\ cm^3 = 72\ L$